

the university's endowment that would have gone to building the dormitory can be allocated to equities, so the dormitory gets built anyway and is used as collateral for the university's muni bond issue. A second example is an individual who receives a tax deduction for the interest on a mortgage. She could borrow using a home equity loan and invest that money in dividend-paying stocks. Money is again fungible. This allows her to allocate the money she is saving for her son's college tuition to equities. Then she borrows money with a home equity loan and uses those proceeds for the money she would have put into her son's college savings fund. One of the largest tax arbitrages (in terms of the cost to the federal government) is the mortgage deduction, which the IRS limits to a mortgage of no more than \$1 million.

4.1. TAX TIMING OPTIONS

In 1983, George Constantinides published a major paper showing how the one-year cut-off between short-term capital gains (which are taxed as income) and long-term capital gains gave rise to tax-timing options. Constantinides showed that it is optimal to realize losses immediately as they arise and to defer (to infinity . . . and beyond!) the realization of capital gains.¹⁹ The loss is deductible at income tax rates, and the deferment delays paying capital gains taxes. Furthermore, the tax strategy is separable from the portfolio choice problem.

There are some strong conditions in Constantinides' original paper, but they have been relaxed in follow-on work.²⁰ For example, in Constantinides' basic model there are no constraints on buying or selling. If there are short sale constraints, then the asset allocation decisions depend on the tax basis (in a complicated fashion). In the original model, there was no limit on tax rebates for capital losses. In the current tax code, only \$3,000 of losses can be offset against ordinary income (but amounts in excess of \$3,000 can be carried over to future tax years). With such a constraint, it may be optimal to realize gains and pay taxes to regain the option of using new losses against other income. But overall the basic idea holds: realize the loss as soon as you can (but with transactions costs it may pay to wait) and try to defer capital gains. So come the end of the tax year, sell your losers.

This advice sounds like the opposite of rebalancing, which I advocated in chapter 4. Rebalancing buys losers. Tax timing sells losers. How are these consistent with each other? The Constantinides tax timing option is different from the underlying asset allocation decision. Suppose there is a 40% allocation to equities, which an investor has implemented using the AA Index Fund. Then at the end

¹⁹Deferring to infinity is not academic. In many circumstances, when you leave appreciated securities to your heirs, the basis is reset at your death.

²⁰See, for example, DeMiguel and Uppal (2005), Gallmeyer and Srivastava (2011), and Marekwica (2012).